

Density Estimation

Recap : $h^*(x) = \operatorname{argmin}_{\mathcal{H}} \left[\mathbb{E}_{P_{X,Y}} L(h(x), y) \right]$

for 0-1 loss, $h_B^*(y) = \begin{cases} 1 & \text{if } P_{y=1|x} > P_{y=0|x} \\ 0 & \text{otherwise} \end{cases}$

Implementing the optimal Bayes classifier demands estimating densities from data.

Eg : Give D , estimate P_X , P_Y , $P_{X|Y}$

↳ class-conditional density.

Maximum likelihood (Minimum KL) Estimation

Given $V = \{v_1, v_2, \dots, v_N\} \sim \text{iid } P_v$,
Estimate P_v .

start with p_0 & $\theta = \operatorname{argmin}_{\Theta} D_{KL}(P_v \parallel P_{\theta})$
 $\equiv \operatorname{argmax}_{\theta} \left[\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(v_i) \right]$
 $v_i \sim \text{iid } P_v$

Examples of ML Estimation :

a) $p_{\theta}(v) \sim N(v; \theta, I)$: Model.

$$v \in \mathbb{R}^d, \quad \theta \in \mathbb{R}^d$$

We have $v_1, v_2, \dots, v_N \sim \text{iid } p_v$.

$$p_{\theta}(v) = \frac{1}{(2\pi)^{d/2} |I|^{1/2}} \exp \left\{ -\frac{1}{2} (v - \theta)^T I (v - \theta) \right\}$$

consider $l(\theta) = \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(v_i)$

$$\propto -\frac{1}{N} \sum_{i=1}^N \|v_i - \theta\|_2^2 = \frac{\partial}{\partial \theta} \left(\sum (v_i - \theta)^T (v_i - \theta) \right) = \sum 2(v_i - \theta)$$

$$\theta^* = \underset{\theta}{\operatorname{argmax}} l(\theta)$$

$$\theta^* = \frac{1}{N} \sum_{i=1}^N v_i$$

Max. like. Estimate

Example : Discrete Random variable

$$D = \{v_1, v_2, \dots, v_N\} \sim \text{iid } p_v$$

$$v_i \in \{a_1, a_2, \dots, a_m\} \text{ with prob. } \{p_1, p_2, \dots, p_m\}$$

parameters $\theta = \{p_1, p_2, \dots, p_m\}$

To express the mass function for this RV, we need one-hot representation.

One hot representation

For each v_i , define an auxiliary RV z_i as follows:

$$z_i = [z_i^1 \ z_i^2 \ \dots \ z_i^m], \quad z_i^j \in \{0, 1\}$$

$$z_i^j = \begin{cases} 1 & \text{pr } j: v_i = a_j \\ 0 & \text{otherwise} \end{cases}$$

eg $D = \{2, 3, 4, 1, 6, \dots\}$
 $v_i \in \{1, 2, 3, 4, 5, 6\}$

$$z_1 = [0 \ 1 \ 0 \ 0 \ 0 \ 0]$$

$$z_2 = [0 \ 0 \ 1 \ 0 \ 0 \ 0]$$

New data $D = \{z_1, z_2, \dots, z_n\}$
 $z_i \in \{0, 1\}^m$

$$p_{\theta}(v_i = a_j) = p_1^{z_i^1} \times p_2^{z_i^2} \dots \times p_m^{z_i^m}$$

$$z_i = \begin{bmatrix} 0 & \dots & 0 & 1 & \dots & 0 \end{bmatrix}$$

$j^{\text{th}}_{\text{pos}}$

$$p_{\theta}(v_i) = \prod_{j=1}^m p_j^{z_i^j}, \quad \theta = \{p_1, p_2, \dots, p_m\}$$

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(v_i)$$

$$= \underset{\theta}{\operatorname{argmax}} \frac{1}{N} \sum_{i=1}^N \log \prod_{j=1}^m p_j^{z_i^j}$$

$$= \underset{p_j}{\operatorname{argmax}} \left[\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^m z_i^j \log p_j \right]$$

adding constraints to the above problem,

$$0 \leq p_j \leq 1 \quad \& \quad \sum_{j=1}^m p_j = 1$$

New objective :

$$\underset{\theta}{\operatorname{argmax}} \quad l(\theta)$$

$$\text{s.t.} \quad 0 \leq p_j \leq 1 \quad \& \quad \sum_{j=1}^m p_j = 1$$

writing the lagrangian,

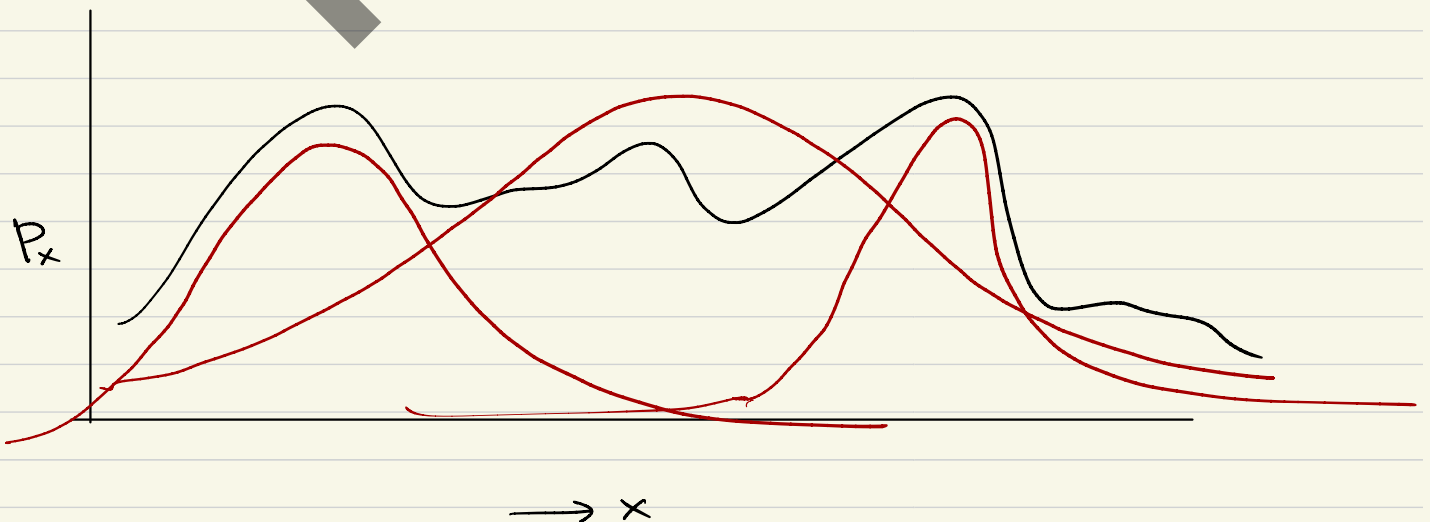
$$\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M z_i^j \log p_j + \alpha \left(\sum_{j=1}^M p_j - 1 \right) + \lambda \left(p_j - 1 \right) + \beta \left(-p_j \right)$$

after solving the above problem,

$$p_j^* = \frac{\text{No. of times } a_j \text{ as occurred}}{N}$$

$$= \frac{\sum_{i=1}^N z_i^j}{N}$$

Density Estimation for Mixture distributions.



Multi-modal densities are not "well-estimated" with uni-modal models.

solⁿ: Try a multi-modal model.

Mixture density function:

Suppose $v \in \mathbb{R}^d$,

Define a mixture density

$$p_{\theta}(v) = \sum_{j=1}^M \alpha_j p_{\theta_j}(v)$$

$\alpha_j \in [0, 1]$ p_{θ_j} : density function.

Gaussian Mixture Model (GMM)

$$p_{\theta_j} \sim \mathcal{N}(\cdot, \theta_j)$$

GMMs are universal density approximators for CRVs.

p_{θ}